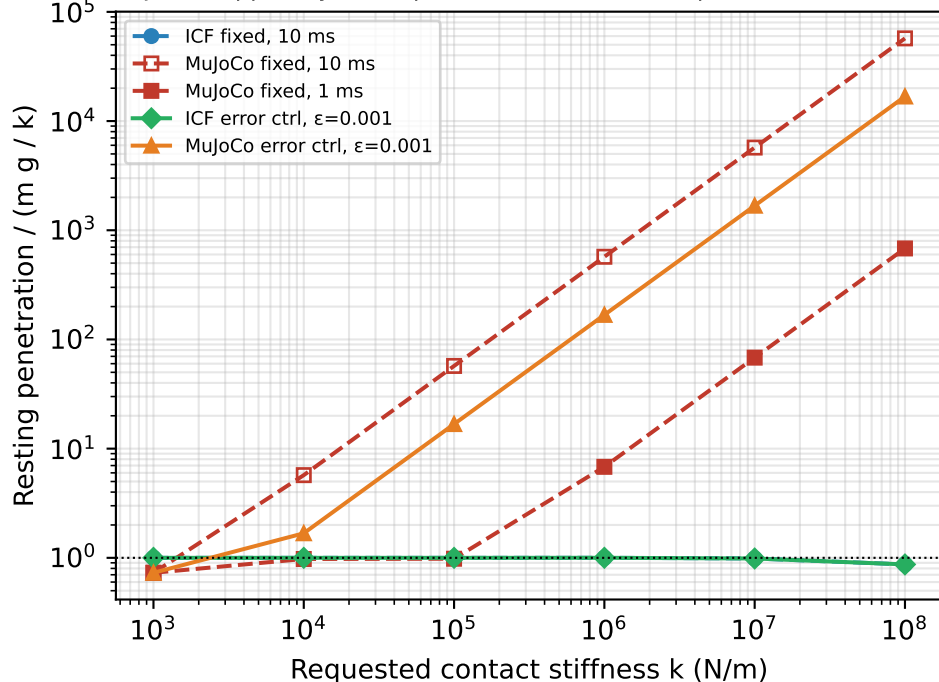
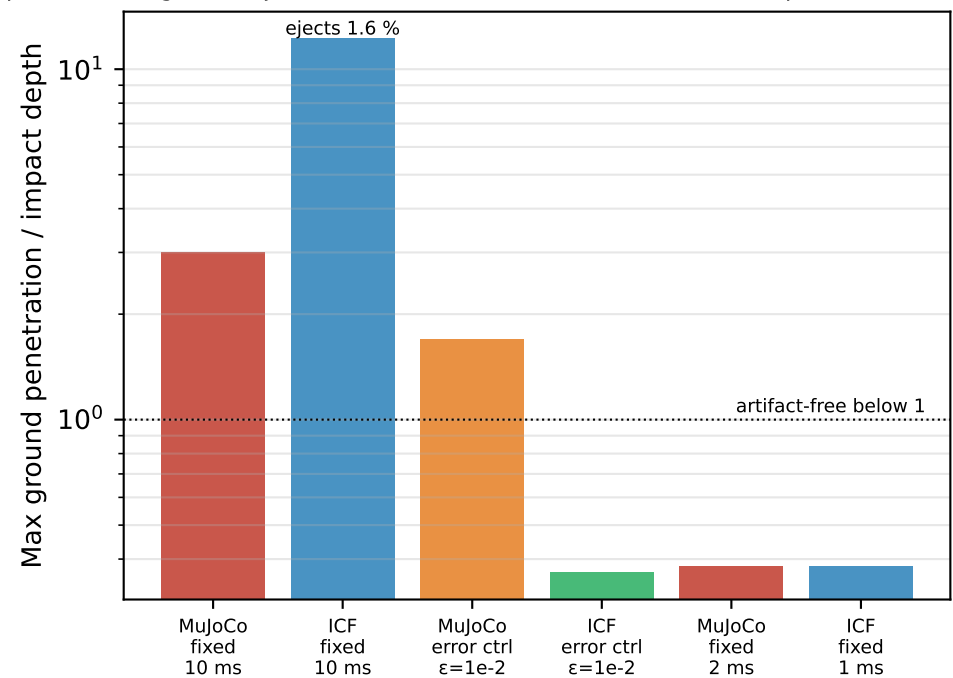


At the step a learner uses, fixed stepping makes artifacts; error control removes them

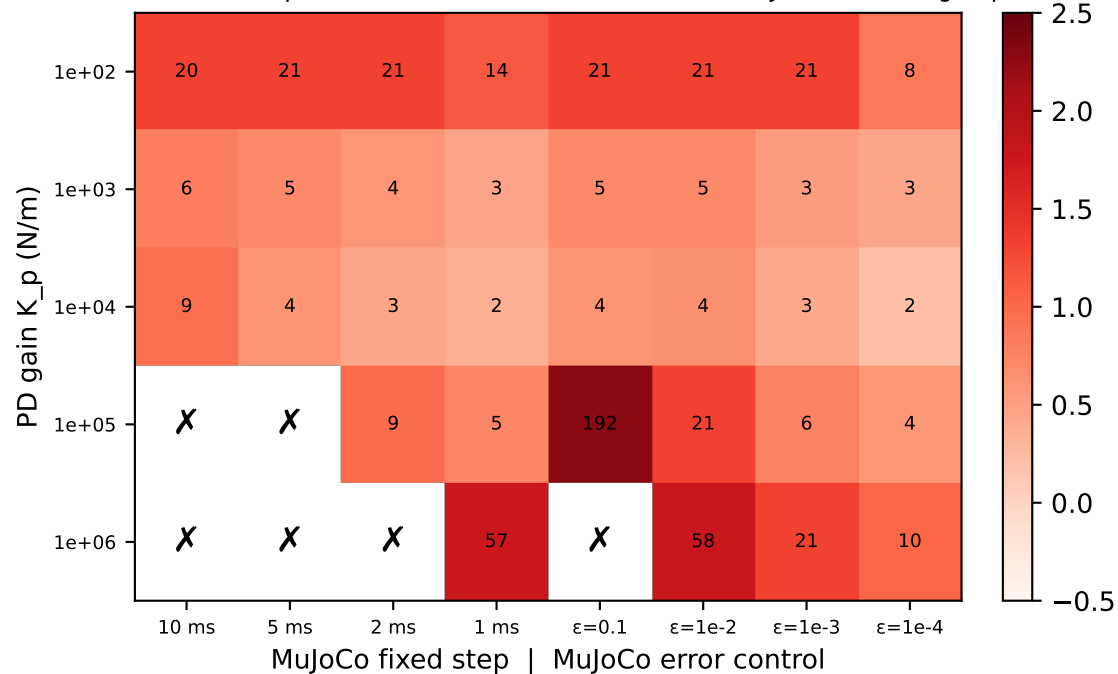
(a) The stiffness each solver realizes: ICF the model at any step; MuJoCo capped by its step ($\tau \geq 2\delta t$), invisible to position error control



(b) Hard clutter at the learner's step (10 ms): both fixed arms are artifacts (ICF passes through and ejects); error control at $\epsilon = 1e-2$ is not; cheapest fixed fixes: 2 ms / 1 ms



(c) A PD gantry pushing a 1 kg box: MuJoCo's box hops (mm) and its joint gain diverges (X) at the learner's steps; ICF: stable in all 40 cells, lift ≤ 0 (stays at its resting depth)



(d) The chatter a held 100 Hz target excites at $K_p = 1e5$ (300 N kicks on a 0.1 kg tip): resolved as the step shrinks, integrated away entirely at 10 ms — a policy never sees it

